RESULTS AND RESEARCH REGISTER FOR SIX JACOBIAN-CONJECTURE PROGRAMS

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Abstract. This companion register preserves results, constructions, open problems, corrections, and research leads associated with six reader manuscripts on the recent three-dimensional Jacobian-conjecture counterexample. It is an index into the research program, not a seventh theorem paper. Each entry states its evidence boundary, and no open, contextual, reported, or evidence-only entry is used as a theorem in the reader manuscripts.

1. How to read this register

The six reader manuscripts contain the selected theorem spines and their proofs. This document records material that is mathematically relevant but would interrupt those narratives. "Proof offered" means that an argument is present in the private evidence archive or a proof-bearing source appendix; it does not mean that independent specialist review has occurred. "Certificate offered" identifies an exact or computer-assisted witness whose precise scope is given in the corresponding computation index.

The primary-home map distributed with version 9 records which reader paper owns each mathematical development and which papers merely use it. The complete version-8 archival PDFs retain the full proof appendices removed from the shorter reader editions.

Part 1. Cubic marked-root covers

2. Supplementary Results and Research Leads

This appendix keeps potentially relevant results from the research archive attached to a mathematical home without enlarging the main theorem spine. This register is published separately from the six reader manuscripts. Inclusion here is deliberately permissive. Each entry states its evidence boundary; entries labelled as open, contextual, or evidence-only are not used as theorems elsewhere in the paper. Earlier formulations known to be false are replaced by their current correction. The computational supplement and its `COMPUTATION.md` index give the available program-level artifact map; an entry here does not turn a shared-code check into an independent verification.

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2.1. **Additional results and constructions.**

Three-dimensional Keller maps of every generic degree. *Status.* Proof offered; independent review is not recorded.
For every integer d at least 3, there is a nonproper Keller map from complex affine 3-space to itself with generic degree d.
Credit. Credit recorded to algal (construction, computation).
Degree-three function-field extension with S3 Galois closure. *Status.* Proof offered; independent review is not recorded.
The function-field extension induced by the Alpöge map has degree 3 and an S3 Galois closure, as described by the cited explicit cubic model.
Credit. Credit recorded to dorky (derivation).
The double-root resultant slice is affine three-space. *Status.* Proof offered; independent review is not recorded.
In the binary-form multiplication model of the counterexample, the resultant-one double-root slice is polynomially isomorphic to affine three-space.
Credit. Credit recorded to Alejandro Radisic (formalization); Andy Jiang (problem suggestion); ChatGPT (research assistance); Daniel Litt (proof); David Speyer (proof); Levent Alpöge (construction); Lillian Ryan Uhl (verification); Ravi Vakil (exposition); Terence Tao (proof, exposition); Will Sawin (proof).
For multiplication-incidence opens from P^a x P^b, the next (2, 3)... *Status.* Proof offered; independent review is not recorded.
For multiplication-incidence opens from P^a x P^b, the next (2, 3) candidate fails to have the affine-space class/count expected of A^5; the original (1, 2) case is singled out within the stated incidence-coordinate class.
A rank-three affine-linear family whose full simple-root incidence... *Status.* Proof offered; independent review is not recorded.
A rank-three affine-linear family whose full simple-root incidence is affine three-space lies in the tangent-but-not-osculating orbit and recovers the base map up to left-right equivalence.
Every generic-degree-three Keller map factors through a finite... *Status.* Proof offered; independent review is not recorded.
Every generic-degree-three Keller map factors through a finite normal cubic cover whose trace-zero module is rank-two reflexive and locally free away from finitely many target points.
Flatness of the finite cubic normalization is equivalent to... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
Flatness of the finite cubic normalization is equivalent to vanishing of a finite defect module expressible through Ext data.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

A minimal defect can only occur over an omitted singleton... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
A minimal defect can only occur over an omitted singleton fiber of scheme length at least four.
The defect problem admits a resolvent maximal-Cohen–Macaulay... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The defect problem admits a resolvent maximal-Cohen–Macaulay reformulation.
At the first possible defect stratum the discriminant has... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
At the first possible defect stratum the discriminant has multiplicity at least six and the exceptional boundary has an elliptic-type constraint.
No algebraically homogeneous minimal defect can occur in a Keller... *Status.* Proof offered; independent review is not recorded.
No algebraically homogeneous minimal defect can occur in a Keller normalization.
A generically degree-three Keller map admits a useful normalization... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
A generically degree-three Keller map admits a useful normalization picture: a finite normalization X over the target together with an open embedding of A3, expressible through a finite-flat binary-cubic model.
The choice A=C(C+1), B=-2-4C produces an exact cubic Keller family... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The choice A=C(C+1), B=-2-4C produces an exact cubic Keller family with an unramified lost sheet, showing that omission need not coincide with ramification.
Within the A, B family, the nonproperness set can have arbitrarily... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
Within the A, B family, the nonproperness set can have arbitrarily many components, with discriminant and unramified-loss components controlled by the roots of explicit coefficient polynomials.
For a finite normalization of the cubic cover, the trace-zero... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
For a finite normalization of the cubic cover, the trace-zero module is reflexive and the nonflat locus is finite; flatness and the global cubic package require an additional Cohen-Macaulay hypothesis.
An affine-root incidence component of degree at least two cannot... *Status.* Proof offered; independent review is not recorded.

An affine-root incidence component of degree at least two cannot be A2, and a generically squarefree affine-linear projective coefficient family has no connected full simple-root incidence isomorphic to A2.
The map is prime under polynomial composition. *Status.* Proof offered; independent review is not recorded.
The map is prime under polynomial composition.
The off-diagonal collision space is a smooth factorial affine... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
The off-diagonal collision space is a smooth factorial affine threefold with trivial Picard group and unit group modulo constants $\mathbb{Z}$.
Escape rates are $\varepsilon^{-1/2}$ near a smooth discriminant... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
Escape rates are $\varepsilon^{-1/2}$ near a smooth discriminant point and $\varepsilon^{-2/3}$ near the cusp, with more degenerate arcs allowing larger half-integral exponents.
Every nonzero constant combination of the inverse-Jacobian vector... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
Every nonzero constant combination of the inverse-Jacobian vector fields is incomplete.
Formal extension of the ruling can hold while Zariski-local... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
Formal extension of the ruling can hold while Zariski-local algebraization of every lift fails; explicit rational threefold and same-incidence nonalgebraizable lifts refute the proposed general algebraization lemma.
The resultant-one factor space is isomorphic to SL2 times A1_gamma... *Status.* Proof offered; independent review is not recorded.
The resultant-one factor space is isomorphic to SL2 times A1_gamma, and its universal marked-root map to the simple-root binary-cubic locus is etale; ordering the roots gives an S3 torsor.
For every generic-degree-three Keller normalization, base change to... *Status.* Proof offered; independent review is not recorded.
For every generic-degree-three Keller normalization, base change to the affine source splits the cubic algebra as S times S[eta]/(eta^2-D), implying flatness at every attained target point and a canonical marked-root form on the source.
If a generic-degree-three Keller map had smooth irreducible... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
If a generic-degree-three Keller map had smooth irreducible nonproperness hypersurface, then it would be surjective, the hypersurface would be isomorphic to A2, and its discriminant double cover would admit a connected etale C3-cover.

For any polynomial potential H(T, c), the marked-root and slope... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
For any polynomial potential H(T, c), the marked-root and slope equations b=r-H_T(t, c), 2a=H(t, c)+tb have Jacobian r/2; composing with t=y+1/x, r=2/x, c=w(x, y)-x^3z has Jacobian -2 whenever the formulas extend polynomially, and generic degree deg_T H.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
On a sheet marked by a simple root t_i of... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
On a sheet marked by a simple root t_i of a cubic, x_i=2/P'(t_i)=2/[A(c)(t_i-t_j)(t_i-t_k)]; collision of the marked root therefore sends that affine source branch to infinity while the finite completion retains the ramification.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
For every coprime normalized cubic-frame pair in the stated family... *Status.* Proof offered; independent review is not recorded.
For every coprime normalized cubic-frame pair in the stated family, the inverse cubic is irreducible with nonsquare discriminant over C(a, b, c), so its generic Galois closure and geometric monodromy are S3.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
Over A(c) nonzero, the normalized cubic discriminant complement is... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
Over A(c) nonzero, the normalized cubic discriminant complement is the product of the punctured c-line and the centered three-point configuration space; if A has s distinct roots, its fundamental group is F_s times B_3 and the permutation monodromy is the standard B_3-to-S3 quotient.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
Modulo triangular target shears, all polynomial root-slope frame... *Status.* Proof offered; independent review is not recorded.
Modulo triangular target shears, all polynomial root-slope frame potentials admit the stated unique torus-weight normal form, with necessary and sufficient layer conditions at weights m<0, m=0, m=1, and m>=2.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
A reduced frame potential of root degree n>=4 has leading... *Status.* Proof offered; independent review is not recorded.
A reduced frame potential of root degree n>=4 has leading coefficient vanishing to order at least floor(n/2)+1 at the retained infinity point; the bound is sharp and yields component degrees (3n+4, 3n+3, 4) for even n and (3n+2, 3n+1, 4) for odd n.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

For every n>=3 the displayed explicit frame potential defines a... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
For every n>=3 the displayed explicit frame potential defines a three-variable polynomial Keller map of generic degree n and determinant -2, attaining the sharp frame-degree bound.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
For every admissible coprime cubic-frame pair, the homogenized... *Status.* Proof offered; independent review is not recorded.
For every admissible coprime cubic-frame pair, the homogenized inverse cubic defines a smooth finite-flat Gorenstein triple cover of affine three-space with trivial rank-two Tschirnhausen bundle and distinguished homogeneous-root frame.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
For the normalized cubic discriminant, with H=3A(c)t+B(c), the... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
For the normalized cubic discriminant, with H=3A(c)t+B(c), the conductor in C[c, t] is H^2 and the conductor ideal in the discriminant ring is generated by B^2-3Ab and 18Aa+Bb; transversely the singularity is the ordinary cusp C[H^2, H^3] inside C[H].
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
Every Keller map has an infinite-dimensional first-order left-right... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
Every Keller map has an infinite-dimensional first-order left-right automorphism space over the dual numbers, naturally identified with polynomial vector fields on the target; stabilization also adds inert diagonal affine automorphisms.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
For positive boundary length, the hidden ordinary complex-point... *Status.* Proof offered; independent review is not recorded.
For positive boundary length, the hidden ordinary complex-point automorphism kernel vanishes and Aut_LR(G_{A, B}) is the finite stabilizer of the decorated boundary scheme (Z_A, B|Z_A).
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
For every coprime admissible cubic-frame pair, the generic... *Status.* Proof offered; independent review is not recorded.
For every coprime admissible cubic-frame pair, the generic function-field extension has no nontrivial deck transformation; ordinary source automorphisms over the identity target are therefore trivial.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

2.2. **Open problems and unresolved assertions.**

A failed general algebraization route. *Status.* Open problem or unresolved assertion.
The proposed general algebraization lemma is false: explicit nonalgebraizable lifts provide counterexamples. A narrower Keller-specific algebraization problem remains open.
Does the U1/U2/B trichotomy hold in the claimed generality beyond... *Status.* Open problem or unresolved assertion.
Does the U1/U2/B trichotomy hold in the claimed generality beyond the normalized family?
Can the common-root integral-closure theorem and left-right... *Status.* Open problem or unresolved assertion.
Can the common-root integral-closure theorem and left-right equivalence result be independently proved or formalized?
The nonsquarefree common-root quotient. *Status.* Open problem or unresolved assertion.
The later proof draft answers the earlier question: the full Artinian class B modulo A is retained at repeated roots. Independent verification remains desirable.

2.3. **Context, corrections, and evidence boundaries.**

Normal cubic covers with S3 monodromy need not be flat... *Status.* Corrected statement; the earlier stronger formulation is not adopted.
Normal cubic covers with S3 monodromy need not be flat: an explicit minimal nonflat normal cubic algebra supplies a countermodel outside the Keller setting.
The remaining minimal defect can be organized by a radial... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The remaining minimal defect can be organized by a radial fibration and an E6-type boundary/discrepancy picture.

Part 2. **Quartic Keller maps**

3. Supplementary Results and Research Leads

This appendix keeps potentially relevant results from the research archive attached to a mathematical home without enlarging the main theorem spine. This register is published separately from the six reader manuscripts. Inclusion here is deliberately permissive. Each entry states its evidence boundary; entries labelled as open, contextual, or evidence-only are not used as theorems elsewhere in the paper. Earlier formulations known to be false are replaced by their current correction. The computational supplement and its `COMPUTATION.md` index give the available program-level artifact map; an entry here does not turn a shared-code check into an independent verification.

3.1. **Additional results and constructions.**

After collision normalization, fixed-degree Keller counterexample... *Status.* Proof offered; independent review is not recorded.
After collision normalization, fixed-degree Keller counterexample searches become finite coefficient schemes; the message gives concrete variable counts for D=4, 5, 6 and recommends exact elimination or modular-to-rational computation.
K_{2, 2} is, after linear elimination, a twisted-cubic affine cone... *Status.* Proof offered; independent review is not recorded.
K_{2, 2} is, after linear elimination, a twisted-cubic affine cone: codimension two with three minimal quadratic generators, so it is not a complete intersection at the vertex.
An ordered-collision incidence construction makes the locus of... *Status.* Proof offered; independent review is not recorded.
An ordered-collision incidence construction makes the locus of generic degree at least d open on the generically finite part; exact generic degree is locally closed and generic degree is lower semicontinuous in this setup.
The degree-four composite-pencil analysis gives a census/reduction... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The degree-four composite-pencil analysis gives a census/reduction and restrictions on primitive-line and rank-one top forms.
If the quartic top form is rank one of pure-fourth-power... *Status.* Proof offered; independent review is not recorded.
If the quartic top form is rank one of pure-fourth-power type H4=v times l^4, the Keller map is an automorphism.
For the triple-linear top form h=x^3 y, the generic charts... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
For the triple-linear top form h=x^3 y, the generic charts reduce to two exceptional divisors denoted Delta_23 and Psi.
Every rank-one quartic map with top form v times l^3... *Status.* Proof offered; independent review is not recorded.
Every rank-one quartic map with top form v times l^3 m in the treated orbit is an automorphism.
The corrected cubic-pencil reduction says that dP wedge dQ wedge... *Status.* Proof offered; independent review is not recorded.
The corrected cubic-pencil reduction says that dP wedge dQ wedge dh=0 forces either a nontrivial fixed component/gcd, a pencil-cube case, or dependence on two linear forms.
The fixed-component Plücker boundary for $h = x^2y^2$ is eliminated. *Status.* Exact computer-assisted proof offered; independent review is not recorded.
For $H_4 = (0, 0, x^2y^2)$ with cubic pencil generated by x^2y and xy^2, the complete high-syzygy solution and lower Keller equations force the third column of the linear part to vanish. This closes the earlier unresolved

fixed-component chart. It is a special rank-one calculation, not a target-span-two theorem.
If the projective image of the quartic leading form H4... *Status.* Proof offered; independent review is not recorded.
If the projective image of the quartic leading form H4 is a curve of degree c, then H4 factors as G times a degree-c binary parametrization evaluated on a coprime source pencil (A, B), with the exact degree relation deg(G)+c deg(A)=4.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
For a quartic leading map of generic Jacobian rank two... *Status.* Proof offered; independent review is not recorded.
For a quartic leading map of generic Jacobian rank two, the kernel of its characteristic Jacobian derivation is the relative algebraic closure of the leading field; the scalar degrees in this field form qZ for a homogeneous period q dividing 4.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
For a quartic Keller deformation of a leading curve, the... *Status.* Proof offered; independent review is not recorded.
For a quartic Keller deformation of a leading curve, the first nonzero normal coefficient below determinant order nine can occur only at an order divisible by the homogeneous period; in the primitive period-four case the first three normal coefficients vanish.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
The normal-resonance theorem extends to every degree d in dimension... *Status.* Proof offered; independent review is not recorded.
The normal-resonance theorem extends to every degree d in dimension three, with determinant threshold 3(d-1); consequently no three-variable Keller map in any degree can have a period-primitive leading line.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
For a quartic Keller map with leading line H4=(P, Q, 0), the... *Status.* Proof offered; independent review is not recorded.
For a quartic Keller map with leading line H4=(P, Q, 0), the first nonzero normal layer Nm is algebraic but not rational over C(P, Q), and the characteristic-field extension has degree at least 4/gcd(4, m).
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
The first-normal analysis reduces quartic leading-line maps to... *Status.* Proof offered; independent review is not recorded.
The first-normal analysis reduces quartic leading-line maps to binary pencils, one quadratic-source composite type, fourth-power or quadratic-square special fibers according to the normal degree, and fixed components subject to explicit valuation divisibility.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
In a quartic leading-line counterexample the first nonzero normal... *Status.* Proof offered; independent review is not recorded.

In a quartic leading-line counterexample the first nonzero normal layer must have degree three: degree one makes the normal coordinate linear, while degree two makes it a quadratic coordinate and reduces the remaining map to an excluded low-degree plane Keller map.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
In the primitive coprime quartic leading-line case, the surviving... *Status.* Proof offered; independent review is not recorded.
In the primitive coprime quartic leading-line case, the surviving normal-degree-three valuation pattern is uniquely the partition 2+1, reducing the leading data to H4=(ell^4, q2^2, 0) and (H3)3=ell q2 with ell not dividing q2.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
No quartic Keller map has leading form z^2(x^2, xy, y^2), despite the... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
No quartic Keller map has leading form z^2(x^2, xy, y^2), despite the period-two binary-cubic resonance allowed in its second normal layer.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
No quartic Keller map has leading form (A^2, AB, B^2) for coprime... *Status.* Proof offered; independent review is not recorded.
No quartic Keller map has leading form (A^2, AB, B^2) for coprime independent ternary quadratic forms A and B.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
No quartic Keller map has either remaining period-one scalar-conic... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
No quartic Keller map has either remaining period-one scalar-conic leading form xy(x^2, xy, y^2) or x^2(x^2, xy, y^2).
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
No quartic Keller map in three variables has a nondegenerate... *Status.* Proof offered; independent review is not recorded.
No quartic Keller map in three variables has a nondegenerate conic as the projective image of its highest homogeneous part.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
No quartic Keller map in three variables has a nondegenerate... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
No quartic Keller map in three variables has a nondegenerate rational cubic as the projective image of its highest homogeneous part.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
A proper basepoint-free rational-quartic leading image of a quartic... *Status.* Proof offered; independent review is not recorded.
A proper basepoint-free rational-quartic leading image of a quartic Keller map cannot be immersed or simply ramified; equivalently, its ramification factor gamma has degree at least two.

Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
If a proper rational-quartic leading image has ramification degree... *Status.* Proof offered; independent review is not recorded.
If a proper rational-quartic leading image has ramification degree two, then its tangent-syzygy module cannot have unbalanced Hilbert-Burch type (1, 3); only type (2, 2) can remain at ramification degree two.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
Projective duality bounds the ramification degree of a proper... *Status.* Proof offered; independent review is not recorded.
Projective duality bounds the ramification degree of a proper rational plane quartic by three. Before the continuation calculation, the earlier Keller exclusions reduced the rational-quartic frontier to $(\deg\gamma,\ \text{tangent-syzygy type}) = (2,(2,2))$ and $(3,(1,2))$; both are now excluded by the rational-quartic frontier theorem in the reader manuscript.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
A proper rational quartic with ramification degree three is... *Status.* Proof offered; independent review is not recorded.
A proper rational quartic with ramification degree three is projectively the tricuspidal quartic, because its dual is a rational nodal cubic; the displayed parametrization has ramification factor xy(x-y) and tangent-syzygy type (1, 2).
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

3.2. **Open problems and resolved research prompts.**

Ordinary total degree 7 is not known to be minimal... *Status.* Open problem or unresolved assertion.
Ordinary total degree 7 is not known to be minimal in dimension three.
The tricuspidal quartic leading-image prompt is resolved. *Status.* Resolved by an exact computer-assisted proof; independent review is not recorded.
The type-$(1,2)$ tangent-syzygy calculation excludes the tricuspidal quartic as the projective leading image of a quartic Keller map, as recorded in the rational-quartic frontier theorem in the reader manuscript.
Credit. Credit recorded to ChatGPT (AI-assisted research proposal).
The balanced ramification-degree-two prompt is resolved. *Status.* Resolved by an exact computer-assisted proof; independent review is not recorded.
The two-root and double-root normal forms exclude the balanced $(2,(2,2))$ tangent-syzygy stratum as a rational-quartic leading image, as recorded in the rational-quartic frontier theorem in the reader manuscript.
Credit. Credit recorded to ChatGPT (AI-assisted research proposal).

3.3. **Context, corrections, and evidence boundaries.**

For fixed n, D, normalized determinant-one polynomial maps form a... *Status.* Reported context; not independently established here.
For fixed n, D, normalized determinant-one polynomial maps form a finite-type coefficient scheme K_{n, D}; its tangent equation at F is the

coefficientwise vanishing of tr((JF)^{-1} JH), equivalently the differential of det J.
Scaling contracts every bounded-degree normalized Keller component...
Status. Reported context; not independently established here.
Scaling contracts every bounded-degree normalized Keller component toward the identity; a component meeting the counterexample locus is generically counterexample, and invariant regular functions are constant along the contraction.
If D_min is the minimum possible maximum ordinary total degree...
Status. Reported context; not independently established here.
If D_min is the minimum possible maximum ordinary total degree of a coordinate of a complex polynomial Keller counterexample in dimension three, then 4 <= D_min <= 7.
Credit. Credit recorded to Angelo Vistoli (attributed contributor); Levent Alpöge (attributed contributor).

Part 3. **Filtered local rigidity**

4. Supplementary Results and Research Leads

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4.1. **Additional results and constructions.**

For the normalized degree-seven map, the exact determinant... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
For the normalized degree-seven map, the exact determinant linearization has 348 variables, rank 327, kernel dimension 21; the normalized affine orbit has dimension 11, leaving ten transverse tangent directions.
In a degree-seven transverse deformation tower, the ten-dimensional...
Status. Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
In a degree-seven transverse deformation tower, the ten-dimensional tangent space reduces to a five-space at second order and a two-plane at third order; all nonzero first-order transverse directions are obstructed by order at most five.

A local standard-basis computation produces pure powers... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
A local standard-basis computation produces pure powers u0^2, u1^4, u2^5, u3^2, u4^5, u5^4, u6^6, u7^3, u8^3, u9^2 in an initial ideal, making it m-primary and yielding set-theoretic formal/analytic isolation of the degree-seven transverse germ.
At degree eight, the determinant linearization has kernel dimension... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
At degree eight, the determinant linearization has kernel dimension 44, hence 33 directions after the same eleven-dimensional affine orbit, and at least five explicit coordinate families occur (three source quadratic z-shears and two target shears).
On the five explicit degree-eight coordinate directions, the... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
On the five explicit degree-eight coordinate directions, the corrected quadratic/cubic equations lift precisely along the pure source-shear or pure target-shear sectors; a residual four-dimensional tangent survives to order three in a larger complement, and the full degree-four modular calculation remains exploratory.
Exact filtered computations for degree bounds E=2, ..., 7 give... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
Exact filtered computations for degree bounds E=2, ..., 7 give tangent/kernel/affine-orbit dimensions; at E=7 the ambient normalized dimension is 348, determinant rank 327, kernel 21, affine image 11, quotient tangent 10, and quadratic obstruction rank 11.
The full normalized affine stabilizer of G is exactly the... *Status.* Proof offered; independent review is not recorded.
The full normalized affine stabilizer of G is exactly the torus diag(tau^-1, tau, tau^2), not merely its identity component.
The residual cubic coefficient w20=lambda in the chosen... *Status.* Proof offered; independent review is not recorded.
The residual cubic coefficient w20=lambda in the chosen birational-frame normal form is entirely affine gauge, with an explicit identity F_lambda = beta_lambda composed with F_0 composed with alpha_lambda.
The exact rational Hilbert-Samuel data through order six are... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
The exact rational Hilbert-Samuel data through order six are (1, 10, 44, 108, 157, 145, 86), with cumulative dimensions (1, 11, 55, 163, 320, 465, 551); exact order seven adds h7=30 and gives cumulative length 581 through order seven.

For any Keller map F, the first-order kernel of the... *Status.* Proof offered; independent review is not recorded.
For any Keller map F, the first-order kernel of the determinant equation consists of JF times divergence-free vector fields; with the correct output filtration the computed first-order quotient vanishes, source reparametrizations are formally trivial, and the calculation stabilizes in the tested range.
The family is formally conjugate coefficient by coefficient while no... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The family is formally conjugate coefficient by coefficient while no conjugacy of uniformly bounded polynomial degree follows.
The literal weight-20 coefficient in the public cubic boundary frame... *Status.* Proof offered; independent review is not recorded.
The literal weight-20 coefficient in the public cubic boundary frame is moved by translation and is therefore gauge rather than a modulus.
The map has a weighted $\mathbb{C}^{\times}$-symmetry. *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The map has a weighted $\mathbb{C}^{\times}$-symmetry.

4.2. **Open problems and unresolved assertions.**
It remains open to certify the intermediate rational Betti numbers... *Status.* Open problem or unresolved assertion.
It remains open to certify the intermediate rational Betti numbers beta_2 through beta_9 of the length-584 Artin algebra and to obtain a genuinely independent second-system reproduction.
Credit. Credit recorded to ChatGPT (AI-assisted formulation).

4.3. **Context, corrections, and evidence boundaries.**
The earlier inference from a restricted five-dimensional slice to... *Status.* Corrected statement; the earlier stronger formulation is not adopted.
The earlier inference from a restricted five-dimensional slice to isolation in the full transverse space is invalid because higher-order arcs can bend out of the slice.
The 656-dimensional six-jet initial quotient. *Status.* Evidence only; this is not asserted as a theorem.
A six-jet initial monomial quotient has length 656 and the recorded Hilbert function. This is not the length of the completed local ring; the current exact calculation gives total length 584.
Earlier modular prediction of length 584. *Status.* Evidence only; this is not asserted as a theorem.
Two-prime modular data predicted the final order-eight contribution and cumulative length 584. Exact rational reconstruction now proves that length, so the modular prediction is retained only as an earlier cross-check, not as the proof.

Part 4. Stable left–right moduli

5. Supplementary Results and Research Leads

This appendix keeps potentially relevant results from the research archive attached to a mathematical home without enlarging the main theorem spine. This register is published separately from the six reader manuscripts. Inclusion here is deliberately permissive. Each entry states its evidence boundary; entries labelled as open, contextual, or evidence-only are not used as theorems elsewhere in the paper. Earlier formulations known to be false are replaced by their current correction. The computational supplement and its `COMPUTATION.md` index give the available program-level artifact map; an entry here does not turn a shared-code check into an independent verification.

5.1. Additional results and constructions.

In the analyzed cubic-incidence families, additional boundary roots... *Status.* Proof offered; independent review is not recorded.
In the analyzed cubic-incidence families, additional boundary roots yield infinitely many stably left-right inequivalent maps, detectable by omitted-boundary components and curve genera.
A collision between two source points persists infinitesimally, and... *Status.* Proof offered; independent review is not recorded.
A collision between two source points persists infinitesimally, and analytically under a small analytic deformation, when the two Jacobians remain invertible.
Over the complex numbers, the omitted set of G_lambda is... *Status.* Proof offered; independent review is not recorded.
Over the complex numbers, the omitted set of G_lambda is exactly the union of the Gamma and Lambda curves, including the p=0 chart.
For generic lambda other than 0 and -324, the unique... *Status.* Proof offered; independent review is not recorded.
For generic lambda other than 0 and -324, the unique positive-genus omitted component and its birational/Albanese data distinguish stable equivalence classes.
The lambda=0 member is stably distinct from the generic members... *Status.* Proof offered; independent review is not recorded.
The lambda=0 member is stably distinct from the generic members by omitted-component count.
At lambda=-324 the boundary polynomial factors, the Lambda locus has... *Status.* Proof offered; independent review is not recorded.
At lambda=-324 the boundary polynomial factors, the Lambda locus has two G_m components, Gamma is P1 minus six points, and the omitted locus has three components; the map still has generic degree three and S3 monodromy.

Within the affine source-target orbit of the public map, total... *Status.* Proof offered; independent review is not recorded.
Within the affine source-target orbit of the public map, total degree seven is minimal.
No target reparametrization of the tested bounded degree at most... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
No target reparametrization of the tested bounded degree at most eight lowers the public map below degree seven.
Every degree-at-most-four map in the specified equivariant ansatz... *Status.* Proof offered; independent review is not recorded.
Every degree-at-most-four map in the specified equivariant ansatz satisfying the Keller equations is tame/an automorphism.
In the primitive linear-rho cubic ansatz, the equations force degree... *Status.* Proof offered; independent review is not recorded.
In the primitive linear-rho cubic ansatz, the equations force degree seven.
The cubic-frame residual coefficient is gauge, every... *Status.* Proof offered; independent review is not recorded.
The cubic-frame residual coefficient is gauge, every degree-at-most-seven member of that framed family is affinely equivalent to the base example, and degree seven is forced there.
Every equivariant Keller map of degree at most five in... *Status.* Proof offered; independent review is not recorded.
Every equivariant Keller map of degree at most five in the analyzed family is an automorphism.
The intersection C(P, Q, R) with polynomials of degree at most six... *Status.* Proof offered; independent review is not recorded.
The intersection C(P, Q, R) with polynomials of degree at most six is spanned by 1, Q, R, implying that no target automorphism of any degree lowers the public map below degree seven.
For the quadratic cubic-frame slice with alpha nonzero... *Status.* Proof offered; independent review is not recorded.
For the quadratic cubic-frame slice with alpha nonzero, q=beta/alpha^2 is a complete invariant under arbitrary polynomial source-target equivalence and stabilization; the alpha=0 line is gauge-equivalent to the base map.
Inside the normalized family, degree-jumping first-order... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
Inside the normalized family, degree-jumping first-order deformations of the base cubic have tangent quotient C[c]/(C plus Cc); imposing degree at most N gives dimension N-1.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
Every quartic frame has T^4 coefficient divisible by c^3 and... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

Every quartic frame has T^4 coefficient divisible by c^3 and first jet satisfying delta+4gamma+12kappa=0; the displayed quartic realizes these conditions with determinant -2, generic degree four, and component degrees (16, 15, 4).
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
For every coprime admissible cubic-frame pair, without assuming A...
Status. Proof offered; independent review is not recorded.
For every coprime admissible cubic-frame pair, without assuming A squarefree, stable left-right equivalence equals ordinary left-right equivalence and is classified by scaling the decorated finite scheme Z_A=Spec C[c]/(A/c) with its unit-valued jet section B|Z_A.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
For every m>=1, the family A=c(1-c)^m and B=-2+4mc+c^2... *Status.* Proof offered; independent review is not recorded.
For every m>=1, the family A=c(1-c)^m and B=-2+4mc+c^2 sum_{j=0}^{m-1}s_jc^j contains an affine-open m-parameter set of pairwise stably inequivalent three-variable Keller maps with fixed generic degree three and fixed component degrees (4m+7, 4m+6, 4).
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
The boundary-data quotient B_N=[U_N/G_m] is a smooth Deligne-Mumford... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The boundary-data quotient B_N=[U_N/G_m] is a smooth Deligne-Mumford stack of dimension 2N-1 whose geometric points classify coprime admissible cubic-frame maps of boundary length N; it is a framed or rigidified boundary stack, not the full unrigidified left-right groupoid.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
If r is a nonzero boundary point with ord_r A=m... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
If r is a nonzero boundary point with ord_r A=m and ord_r B=n>0, the largest power of c-r dividing the cubic discriminant is (c-r)^{min(m, 2n)}.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
The displayed two-parameter quartic-frame family G_{rho, sigma}...
Status. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The displayed two-parameter quartic-frame family G_{rho, sigma} consists of polynomial Keller maps with determinant -2, generic degree four, and component degrees (16, 15, 4).
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
Within the quartic family, stable left-right equivalence holds... *Status.* Proof offered; independent review is not recorded.
Within the quartic family, stable left-right equivalence holds exactly when (rho, sigma) agrees; the normalized residual discriminant has intrinsic cusp

and double-double strata and conductor divisor 2L+Q, so the family gives an affine plane of fixed-degree stable moduli.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

5.2. **Context, corrections, and evidence boundaries.**

The recorded degree-five and degree-six finite-field scans find no... *Status.* Evidence only; this is not asserted as a theorem.
The recorded degree-five and degree-six finite-field scans find no candidates in their enumerated strata, and the degree-six F5 candidates have no recorded lifts mod 25.

Part 5. **Homogeneous descendants**

6. Supplementary Results and Research Leads

This appendix keeps potentially relevant results from the research archive attached to a mathematical home without enlarging the main theorem spine. This register is published separately from the six reader manuscripts. Inclusion here is deliberately permissive. Each entry states its evidence boundary; entries labelled as open, contextual, or evidence-only are not used as theorems elsewhere in the paper. Earlier formulations known to be false are replaced by their current correction. The computational supplement and its `COMPUTATION.md` index give the available program-level artifact map; an entry here does not turn a shared-code check into an independent verification.

6.1. **Additional results and constructions.**

An explicit degree-three counterexample in 11 variables. *Status.* Proof offered; independent review is not recorded.
The cited map from complex affine 11-space to itself has total degree 3, 52 nonzero monomial terms, constant Jacobian determinant -2, and three displayed distinct rational inputs with one common image.
Credit. Credit recorded to ChatGPT (research assistance); Spacerat (computation).
An explicit cubic-homogeneous counterexample in 24 variables. *Status.* Proof offered; independent review is not recorded.
The cited explicit map has the form G(U)=U+H(U) over Q^24, with every nonzero component of H homogeneous cubic, determinant 1, 54 nonzero cubic monomials, and a displayed collision of two distinct rational points.
Credit. Credit recorded to William Thompson (computation).
Starting from the public 11D counterexample, rank-sensitive cubic... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
Starting from the public 11D counterexample, rank-sensitive cubic suspension yields an explicit 19D cubic-homogeneous Keller counterexample with collision.

An explicit square-zero Druzkowski pairing of the 19D cubic gives... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

An explicit square-zero Druzkowski pairing of the 19D cubic gives a 135D Druzkowski counterexample with collision.

The earlier 26D cubic-homogeneous and 158D Druzkowski constructions... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The earlier 26D cubic-homogeneous and 158D Druzkowski constructions remain valid explicit examples but are superseded as upper bounds by the later 19D and 135D examples.

For an n-dimensional degree-at-most-three map whose quadratic span... *Status.* Proof offered; independent review is not recorded.

For an n-dimensional degree-at-most-three map whose quadratic span has rank r, a block-determinant suspension produces a cubic-homogeneous map in dimension n+r+1.

For the 19D example, the generic Jacobian perturbation JH has... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

For the 19D example, the generic Jacobian perturbation JH has Jordan type (18, 1); nonzero scalar fibers are conjugate and the zero fiber is an automorphism degeneration.

The explicit constructions give the upper bound N_min <= 19... *Status.* Proof offered; independent review is not recorded.

The explicit constructions give the upper bound N_min <= 19 for cubic-homogeneous counterexamples; the conversation uses the elementary no-collinear-collision argument for the lower bound N_min >= 5.

The five-attacks package constructs a 38D Hessian quartic with... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The five-attacks package constructs a 38D Hessian quartic with Hessian Jordan type (35, 2, 1) and derives exact central-binomial inverse-ray coefficients with infinitely many nonzero Laplacian terms in the stated parity class.

Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).

The explicit pairing dictionary gives a 110D Druzkowski... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The explicit pairing dictionary gives a 110D Druzkowski construction, while the derivative-space zero-block and equality-case argument give the model-relative bound 52 <= pairing rank <= 110 for the fixed 19D tensor.

The unique scalar-monomial quadratic shift lowering the cubic jet... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The unique scalar-monomial quadratic shift lowering the cubic jet from rank seven to six is P2=-d^2 e_a, and its quartic error is detected by an exact 13-term functional outside the cubic homological image; the natural quadratic target cleanup also fails.
An exact row-defect theorem holds in the two-matrix square-zero... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
An exact row-defect theorem holds in the two-matrix square-zero family; the accompanying finite-field search found no better example in the tested range.
A second counterexample $G : \mathbb{C}^4 \to \mathbb{C}^4$ has... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
A second counterexample $G : \mathbb{C}^4 \to \mathbb{C}^4$ has determinant 4 and generic degree 8.
A straightforward explicit reduction produced a cubic-homogeneous... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
A straightforward explicit reduction produced a cubic-homogeneous map in 79 variables and a Drużkowski cubic-linear map in 426 variables.
MC($SU(79)$) is false. *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
MC($SU(79)$) is false.
The new 24-variable map improves this to MC($SU(N)$)... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The new 24-variable map improves this to MC($SU(N)$) false for all $N \geq 24$.
The rank-sensitive cubic suspension is triangularly left-right... *Status.* Proof offered; independent review is not recorded.
The rank-sensitive cubic suspension is triangularly left-right equivalent to (X, v, t) mapped by (t^{-1}K(tX), v, t); it preserves generic degree and the geometric Galois closure after adjoining transcendental variables.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
For any Keller map F, the symmetric double (F(x), JF(x)^T y)... *Status.* Proof offered; independent review is not recorded.
For any Keller map F, the symmetric double (F(x), JF(x)^T y) is polynomially right-equivalent to F times the identity, so the 38-variable gradient descendant has the same generic degree and monodromy as the 19-variable cubic map.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
A full-rank square-zero Gorni-Zampieri pairing H=B(Dz)^{*3}, BD=0... *Status.* Proof offered; independent review is not recorded.
A full-rank square-zero Gorni-Zampieri pairing H=B(Dz)^{*3}, BD=0, is triangularly right-equivalent to the paired cubic map times an identity

factor; the 110-variable Druzkowski map is therefore a stable presentation of the 19-variable cubic map.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
The original three-variable map has the displayed invariant quotient...
Status. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The original three-variable map has the displayed invariant quotient cubic theta^3-2theta^2+Btheta-2A, whose irreducibility and nonsquare discriminant give generic Galois group S3; the descendant ladder presents this same three-sheeted cover.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
For the fixed 38-variable Hessian quartic, Delta^m(Q^{m+1}) is... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
For the fixed 38-variable Hessian quartic, Delta^m(Q^{m+1}) is nonzero for every m>=1; an explicit two-parameter inverse ray removes the earlier parity gap and ties every nonzero coefficient to a sheet escaping at the discriminant.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
Although every JH(z) is nilpotent, the product JH(e_d)JH(e_t) has...
Status. Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
Although every JH(z) is nilpotent, the product JH(e_d)JH(e_t) has characteristic polynomial lambda^18(lambda+3); hence the polynomial-matrix family is not strongly nilpotent and has no common constant triangularizing basis.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
The equivariant cubic rank-six compression condition has a... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The equivariant cubic rank-six compression condition has a 20-dimensional affine slice, and the quartic obstruction functional is identically one on that slice; modulo two target-row directions this is the natural transverse compression space.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
The n+r+1 cubic suspension is coordinate-free on V plus the... *Status.* Proof offered; independent review is not recorded.
The n+r+1 cubic suspension is coordinate-free on V plus the dual of the output-mode span plus one scalar, and r=rank(C^flat) is the minimum possible auxiliary dimension for any factorization of the cubic tensor through an intermediate vector space.
Credit. Credit recorded to ChatGPT (AI-assisted derivation and drafting).
Within the fixed 38-dimensional symmetric double, there is no proper...
Status. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

Within the fixed 38-dimensional symmetric double, there is no proper nondegenerate invariant linear slice surjecting onto the original variables; any smaller realization must change the realization mechanism rather than delete such a slice.
Credit. Credit recorded to ChatGPT (AI-assisted critique and derivation).
If an n-dimensional homogeneous map has a k-dimensional... *Status.* Proof offered; independent review is not recorded.
If an n-dimensional homogeneous map has a k-dimensional partial-gradient block, it admits a weak symmetric dilation in dimension at most 2n-k; in particular every polynomial map has weak symmetric-dilation dimension at most 2n-1.
Credit. Credit recorded to ChatGPT (AI-assisted critique and derivation).
For the displayed 19-dimensional block tensor... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
For the displayed 19-dimensional block tensor H(X, w, t)=(tQ(X)+t^2Bw, -q(X), 0), the weak symmetric-dilation invariant is at most 37-dim(w), hence at most 36.
Credit. Credit recorded to ChatGPT (AI-assisted critique and derivation).
The admissible symmetric-overlap index kappa gives the lower bound... *Status.* Proof offered; independent review is not recorded.
The admissible symmetric-overlap index kappa gives the lower bound sigma_sym(H)>=2n-kappa_ov(H); when the overlap form is nondegenerate, the partial-gradient construction gives the matching upper bound 2n-kappa_nd(H).
Credit. Credit recorded to ChatGPT (AI-assisted critique and derivation).
In the scalar-w block case, a 35-dimensional weak dilation extending... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
In the scalar-w block case, a 35-dimensional weak dilation extending the evident (w, t)-compression exists exactly when there is a nonzero vector a satisfying a^TB=0, a^TQ(X)=0, and D_a q(X)=0 identically.
Credit. Credit recorded to ChatGPT (AI-assisted critique and derivation).
For the principal sl2 conic on Sym^m(k^2), the integrability... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
For the principal sl2 conic on Sym^m(k^2), the integrability operator has determinant -m^{m+2}(m+2)^m; no direct sum of principal conics supports a noncollinear normalized collision.
Credit. Credit recorded to ChatGPT (AI-assisted critique, derivation, and computation).
An everywhere-regular five-dimensional quadratic nilpotent line... *Status.* Proof offered; independent review is not recorded.
An everywhere-regular five-dimensional quadratic nilpotent line pencil whose kernel map spans P4 has saturated filtration degrees (-4, -2, 0, 2, 4);

every successive map is an isomorphism and the kernel map is the rational normal quartic.
Credit. Credit recorded to ChatGPT (AI-assisted critique, derivation, and computation).
Every quadratic matrix annihilating the Veronese kernel k_m factors...
Status. Proof offered; independent review is not recorded.
Every quadratic matrix annihilating the Veronese kernel k_m factors uniquely as N=L_mS_m; iterated Hilbert-Burch descent preserves regular nilpotence, and a fully compatible canonical descent chain is necessarily a common scalar multiple of the principal chain.
Credit. Credit recorded to ChatGPT (AI-assisted critique, derivation, and computation).
With the quartic kernel fixed at the principal point, the... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
With the quartic kernel fixed at the principal point, the nilpotent-pencil tangent space has dimension 16 and sl2 type V0+V2+V4+V6; fixing the descended cubic kernel leaves only scalar rescaling, so the other 15 directions are exactly relative descended-cubic motion.
Credit. Credit recorded to ChatGPT (AI-assisted critique, derivation, and computation).
For a general five-dimensional quadratic pencil, integrable markings...
Status. Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
For a general five-dimensional quadratic pencil, integrable markings lie on the proper discriminant det Phi_N=0, and normalized collisions lie on the determinantal locus rank Theta_N<=9 together with the open independence condition on the recovered marking vectors.
Credit. Credit recorded to ChatGPT (AI-assisted critique, derivation, and computation).
For a global regular-type five-dimensional nilpotent quadratic... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
For a global regular-type five-dimensional nilpotent quadratic tensor with full-span collision-line kernel, the saturated rank-one defects are nested and have total codimension-two class 20H^2, combined Hilbert polynomial 10n^2+50n+81, and weighted effective increments 5Delta_1+4Delta_2+3Delta_3+2Delta_4+Delta_5=20H^2.
Credit. Credit recorded to ChatGPT (AI-assisted critique, derivation, and computation).

6.2. Open problems and unresolved assertions.

What is the true smallest dimension of a cubic-homogeneous... *Status.* Open problem or unresolved assertion.

What is the true smallest dimension of a cubic-homogeneous counterexample?
What is the exact minimal square-zero Druzkowski pairing rank for...
Status. Open problem or unresolved assertion.
What is the exact minimal square-zero Druzkowski pairing rank for the 19D cubic?

6.3. **Context, corrections, and evidence boundaries.**

Claimed failure of the Mathieu conjecture for SU(3). *Status.* Correction retained: the original positive claim is not adopted; only the displayed negative scope statement is current.
The cited dimension-varying implication does not establish that the Mathieu conjecture for the fixed group SU(3) is false.
Credit. Credit recorded to Zihan Zhang (derivation, exposition).
Within a zero-diagonal five-by-five sparse matrix ansatz with at... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
Within a zero-diagonal five-by-five sparse matrix ansatz with at most four nonzero entries, no Keller map in the specified H_C family realizes the normalized collision.
A second issue occurs in Zwart's fixed-highest-weight-vector proof. *Status.* Reported context; not independently established here.
A second issue occurs in Zwart's fixed-highest-weight-vector proof.
No explicit Mathieu witness pair is currently recorded. *Status.* Reported context; not independently established here.
The Mathieu argument currently proves existence of a violating pair, but no explicit closed-form pair of functions on SU(24) or SU(79) is recorded. Making the generic Cartan-twist parameter explicit remains open.
The direct Euclidean 36-dimensional partial-gradient potential is... *Status.* Corrected statement; the earlier stronger formulation is not adopted.
The direct Euclidean 36-dimensional partial-gradient potential is not Keller when B is nonzero; obtaining a 36-dimensional Keller or cover-preserving realization is a separate self-adjoint nilpotent matrix-completion problem.
Credit. Credit recorded to ChatGPT (AI-assisted critique and derivation).

Part 6. **Plane boundary obstructions**

7. Supplementary Results and Research Leads

This appendix keeps potentially relevant results from the research archive attached to a mathematical home without enlarging the main theorem spine. This register is published separately from the six reader manuscripts. Inclusion here is deliberately permissive. Each entry states its evidence boundary; entries labelled as open, contextual, or evidence-only are not used as theorems elsewhere in the paper. Earlier formulations known to be false are replaced by their current correction.

The computational supplement and its `COMPUTATION.md` index give the available program-level artifact map; an entry here does not turn a shared-code check into an independent verification.

7.1. **Additional results and constructions.**
Directly slicing or graph-pulling back the three-dimensional cubic... *Status.* Proof offered; independent review is not recorded.
Directly slicing or graph-pulling back the three-dimensional cubic mechanism does not produce a nontrivial plane Keller counterexample: fixed nonzero slices are G_m x A1, the zero slice splits off a trivial A2 sheet, degree-two/three A2 components are excluded, and the one-pole polynomial lift cannot have constant Jacobian.
At a target DVR, the invariant discriminant exponent is sum... *Status.* Proof offered; independent review is not recorded.
At a target DVR, the invariant discriminant exponent is sum f_i(e_i-1), while a monic integral primitive polynomial has exponent sum f_i(e_i-1)+2 iota_C; a reduced primitive discriminant component therefore forces one (e, f)=(2, 1) ramified divisor and zero monogenic index.
The finite-dicritical Newton-Puiseux chart satisfies an exact... *Status.* Proof offered; independent review is not recorded.
The finite-dicritical Newton-Puiseux chart satisfies an exact coefficient recurrence; an exact two-level expansion forces a smooth embedded A1 component and hence cannot occur for a plane Keller counterexample.
The proposed finite-order three-level 2:3 Puiseux rigidity lemma is... *Status.* Proof offered; independent review is not recorded.
The proposed finite-order three-level 2:3 Puiseux rigidity lemma is false: for every immersed limit parametrization the recurrence extends at each finite order, and the nodal model (xi^2, xi^3-xi) has an explicit all-orders algebraic-series solution.
The initial theorem claiming all finite quadratic-parameter... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The initial theorem claiming all finite quadratic-parameter incidence surfaces are of log general type was overbroad because ramification over parameter infinity overlaps the reduced boundary. The retained theorem requires squarefree behavior at infinity or the corrected log-pair hypotheses.
A scratch calculation for Borisov's First Framework derives cycle... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
A scratch calculation for Borisov's First Framework derives cycle type 5 1^11, invariant discriminant order four, primitive exponent 4+2 iota, and provisional curve invariants C^2=13 and arithmetic genus seven.

The two surviving normalized (8, 28) Newton alternatives are encoded... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The two surviving normalized (8, 28) Newton alternatives are encoded as exact saturated sparse coefficient ideals: 72 variables/92 nonzero bracket equations in the truncated case and 186 variables/302 equations in the full case.
The upper face has common approximate root R=y^7(y-1); the exact... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The upper face has common approximate root R=y^7(y-1); the exact layer cokernels persist, and the first nonlinear compatibility condition forces the first correction to be tangent to the locus (R^2, R^3), rather than yielding a contradiction.
The lower face of either Newton alternative forces a degree-21... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The lower face of either Newton alternative forces a degree-21 Belyi map tau=z q^2/p^3 with passport (2^10 1), (3^7), (17 1^4).
There are exactly five connected dessin isomorphism classes with the... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
There are exactly five connected dessin isomorphism classes with the forced degree-21 passport.
The forced residue degree 21 conflicts with the degree-16 map... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The forced residue degree 21 conflicts with the degree-16 map on Borisov's published principal -5 component, conditionally excluding Three-dessin if the toric and Borisov divisors are intrinsically the same valuation.
The five degree-21 dessins form one degree-five Galois orbit over... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The five degree-21 dessins form one degree-five Galois orbit over Q, described by one explicit irreducible quintic field and rational reconstruction formulas for p and q.
For the full polygon, all b=0 branches and the ordinary... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
For the full polygon, all b=0 branches and the ordinary b=+/-1, a=0 or 1 branches are eliminated exactly over the quintic field.
Both exceptional full-polygon four-variable ideals are unit ideals... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.

Both exceptional full-polygon four-variable ideals are unit ideals over the specified quintic field, with explicit exact Nullstellensatz certificates and independent exact replay.

The degree-21 lower-face equation has exactly five explicit... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.

The degree-21 lower-face equation has exactly five explicit normalized solutions forming one quintic Galois orbit, and the five dessins have monodromy A21.

The vertex-saturated truncated Newton support is empty in... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.

The vertex-saturated truncated Newton support is empty in characteristic zero for every one of the five lower faces.

Earlier modular stage of the full-support exclusion. *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The full-support reduction gives an exact system of fifteen equations in five variables over the quintic field. The original modular unit-ideal check has since been replaced by exact characteristic-zero certificates.

The six selected full-support obstruction equations generate the... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The six selected full-support obstruction equations generate the unit ideal over the quintic field, certified by a specified 7121-by-7121 exact minor whose reduction has determinant 859 modulo 2053.

Credit. Credit recorded to ChatGPT (AI-assisted derivation, computation, and drafting).

Both encoded sparse-support alternatives are empty after... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.

Both encoded sparse-support alternatives are empty after substitution of the degree-21 lower faces.

The full-support layer equations define intrinsic obstruction... *Status.* Proof offered; independent review is not recorded.

The full-support layer equations define intrinsic obstruction classes in cokernels of differential operators on the marked lower boundary curve; the exact unit ideal is the nonvanishing of a Newton-bounded boundary-gluing Kuranishi section.

After a canonical target shear and a (4, 17)-weighted target blowup... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

After a canonical target shear and a (4, 17)-weighted target blowup, the adjacent upper boundary component carries a forced degree-17 Belyi map with passport (4^4 1), (17), (5 1^12).

Five selected partial-gluing equations have mixed volume 296, and... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
Five selected partial-gluing equations have mixed volume 296, and the sixth obstruction has a nonzero norm on the certified 296-dimensional modular finite algebra; across the five split embeddings the rational norm is 51 modulo 2053.
A finite étale torus special fiber of length equal to... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.
A finite étale torus special fiber of length equal to the mixed volume exhausts the complete toric intersection and lifts to a finite flat degree-296 partial-gluing scheme.
Credit. Credit recorded to ChatGPT (AI-assisted derivation, computation, and drafting).
The reduced degree-21 and degree-17 boundary covers are mutually... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The reduced degree-21 and degree-17 boundary covers are mutually compatible, but none of their 296 virtual partial normal-jet gluings extends to a Newton-bounded Keller surface map.
For either surviving Newton support, the (-2, 1) source valuation... *Status.* Proof offered; independent review is not recorded.
For either surviving Newton support, the (-2, 1) source valuation dominates Borisov's target F__-5 with normal ramification index one and residue degree 21, excluding Borisov's Three-dessin framework for those supports.
Any plane counterexample of maximum coordinate degree below 125 has... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
Any plane counterexample of maximum coordinate degree below 125 has generic degree at least 21.
The degree-21 passport has exactly five normalized Belyi maps; they... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The degree-21 passport has exactly five normalized Belyi maps; they are reconstructed over one irreducible quintic field, form one arithmetic orbit with signature (1, 2), and have monodromy A21.
A primitive monomial Newton-face Jacobian equation yields an exact... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
A primitive monomial Newton-face Jacobian equation yields an exact logarithmic derivative; under the stated coprimality and endpoint hypotheses the induced rational map is Belyi and its passport is determined by the face exponents.

For the infinite F2 complete-chain family, the first face has... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

For the infinite F2 complete-chain family, the first face has exactly two normalized solutions with rigid degree-three and degree-six dessins, and the exact-power lift P=S^m is impossible for every m>=2 by an odd-versus-even divisor-valuation obstruction.

The normal-boundary determinant equation has the displayed universal... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The normal-boundary determinant equation has the displayed universal order-r linear deformation operator and residue adjoint, giving an intrinsic residue interpretation of the coefficient compatibility conditions.

Contact data (a, e) with e>a^2 force an explicit hypergeometric... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

Contact data (a, e) with e>a^2 force an explicit hypergeometric secondary boundary polynomial and a degree-e Belyi map with passport (a^a, e-a^2), (e), (a+1, 1^(e-a-1)).

The complete-chain lattice gap quotients terminal primary covers to... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The complete-chain lattice gap quotients terminal primary covers to a finite Hurwitz problem; through maximum degree 128 the explicit reduced covers are isolated and infinitesimally rigid modulo source rescaling.

For the F2 chain (5, 20)->(7/5, 2) with (m, n)=(3, 5), exact shear... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

For the F2 chain (5, 20)->(7/5, 2) with (m, n)=(3, 5), exact shear propagation gives 4433 and 12340 support positions, explicit C5-equivariant two-point windows, and the first missing linearized forcing coefficient at normal order 510.

The apparent F2 external conditions at orders 510 and 520... *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.

The apparent F2 external conditions at orders 510 and 520 are slice-dependent and can both be cancelled exactly over Q by later kernel parameters; neither is a gauge-invariant terminal obstruction.

The normal-layer operator has the logarithmic-connection form... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The normal-layer operator has the logarithmic-connection form D_r(a, b)=AB nabla_r eta+((alpha+beta-r)/(alpha+beta))xi Omega and extends to a gauge complex whose resonant layer is an exact-differential problem.

Credit. Credit recorded to ChatGPT (AI-assisted derivation, computation, and drafting).

For finite Newton/pole windows, the dual of the cokernel of... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

For finite Newton/pole windows, the dual of the cokernel of the layer operator is represented by filtered principal parts satisfying the projected formal-adjoint conditions, and every coefficient-matrix left-null vector converts canonically to the corresponding residue functional.

Credit. Credit recorded to ChatGPT (AI-assisted derivation, computation, and drafting).

The exact layer replay identifies all fifteen archived normal-jet... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.

The exact layer replay identifies all fifteen archived normal-jet equations with filtered residue coordinates; the five layer-seven and six layer-eight tuples are complete pole-filtered frames, while the selected six equations are a decisive but not presently canonical coordinate projection.

Credit. Credit recorded to ChatGPT (AI-assisted derivation, computation, and drafting).

7.2. **Open problems and unresolved assertions.**

Natural next problems include classification of generic-degree-three... *Status.* Open problem or unresolved assertion.

Natural next problems include classification of generic-degree-three examples, bounded-degree local moduli, minimum coordinate degree, nonproperness divisors, optimized cubic reduction, the plane case, and possible universality of the marked-root construction.

Eliminating both normalized (8, 28) alternatives gives the... *Status.* Open problem or unresolved assertion.

Eliminating both normalized (8, 28) alternatives gives the conditional lower bound max(deg P, deg Q)>=125 for any hypothetical plane Keller counterexample, subject to replaying and auditing the upstream Newton reduction.

Exceptional Full-Polygon Ideals Are Unit Ideals. *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.

The two exceptional full-polygon ideals contain 1 over the quintic coefficient field, as witnessed by the exact terminal certificates.

Characteristic-Zero Nullstellensatz Certificates. *Status.* Exact or computer-assisted certificate offered; see the computation index for its scope and independence boundary.

Explicit characteristic-zero Nullstellensatz certificates for the two exceptional full-polygon ideals have been generated and replayed independently in two exact implementations.

Is the toric -5 divisor intrinsically Borisov's principal -5 divisor? *Status.* Open problem or unresolved assertion.
Is the toric -5 divisor intrinsically Borisov's principal -5 divisor?
Can the all-orders nodal local solution be ruled out by... *Status.* Open problem or unresolved assertion.
Can the all-orders nodal local solution be ruled out by a finite-support or global one-double-point theorem?
Top priorities are independent certificate reproduction, the full... *Status.* Open problem or unresolved assertion.
Top priorities are independent certificate reproduction, the full bounded-degree local ring and degree-growth interface, intrinsic triple-cover defect exclusion, boundary-complete rigidity, the minimum degree-three coordinate bound, improved descendant dimensions/tensor rank, the five-dimensional classification, and explaining the two-dimensional obstruction.
Recover the six decisive obstruction components canonically from the... *Status.* Open problem or unresolved assertion.
Recover the six decisive obstruction components canonically from the residue pole filtration or the determinant of cohomology.
Uniqueness of the degree-21 valuation remains open. *Status.* Open problem or unresolved assertion.
It remains open whether the identified source divisor is the only divisor above the generic point of the target F-minus-5 component. The present result gives generic degree at least 21, not equality.
The minimal ordinary degree in dimension three, minimal... *Status.* Open problem or unresolved assertion.
The minimal ordinary degree in dimension three, minimal homogeneous-reduction dimension, and minimal support remain open.

7.3. **Context, corrections, and evidence boundaries.**

The highest-leverage program is exact reconstruction of the five... *Status.* Recorded result or structural lead; it is not promoted beyond the evidence described here and in the project archive.
The highest-leverage program is exact reconstruction of the five Belyi maps followed by weight-filtered tests against the two surviving Newton supports, with independent certificate infrastructure in parallel.
Earlier finite-field support evidence. *Status.* Evidence only; this is not asserted as a theorem.
The finite-field layer-five-through-seven calculation was an early cross-check. It is superseded as evidence by the exact characteristic-zero full-support certificate.
The public degree-below-125 theorem. *Status.* Proof offered; independent review is not recorded.
The public degree-below-125 theorem establishes that every plane Keller map with maximum coordinate degree below 125 is an automorphism.

This manuscript credits that public theorem and does not claim priority; the earlier internal derivation was incomplete.

Disclosure

AI systems were used extensively in exploration, proof drafting, symbolic program development, source organization, and manuscript editing. They are not authors. Entries are governed by their displayed evidence boundaries; human specialist review has not been completed.